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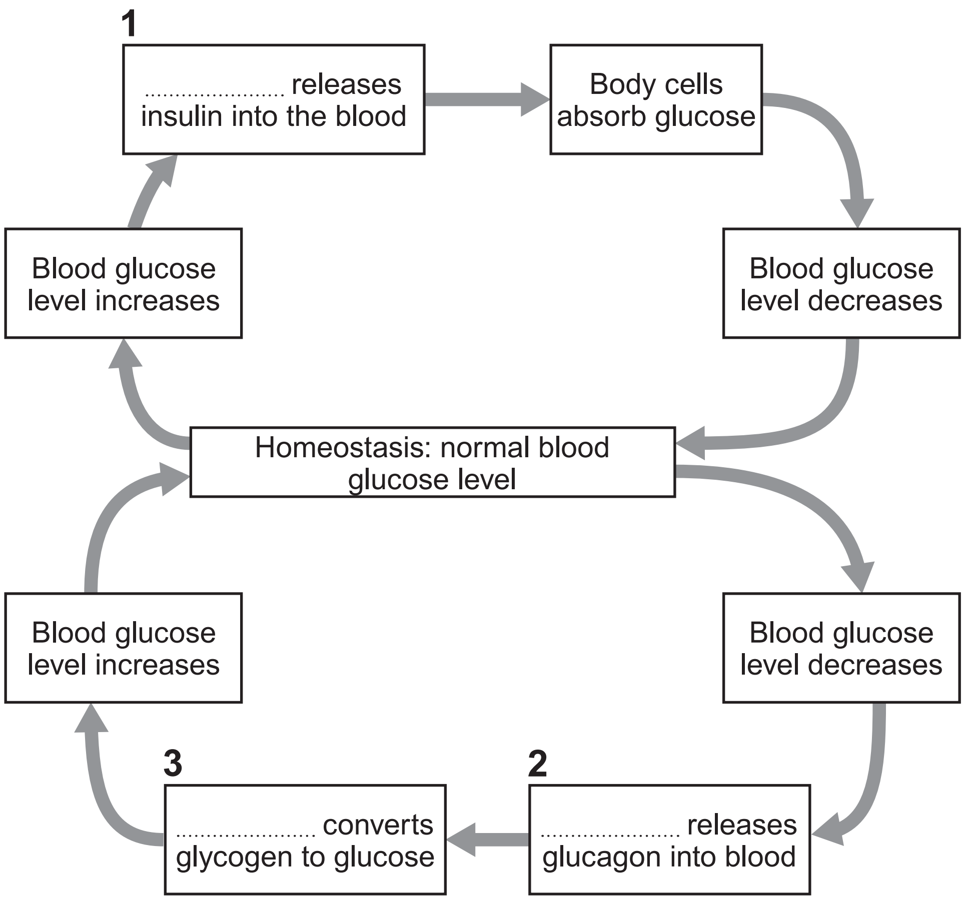
3. **Image 3.1** shows the homeostatic control of glucose levels in the blood of a person.

Image 3.1



(a) (i) **Complete the information** in boxes **1**, **2** and **3** in **Image 3.1** by writing in the name of the **organ** responsible. [3]

(ii) State the meaning of the term homeostasis. [1]

....

....

(iii) Insulin and glucagon belong to a group of chemicals which are produced in one part of the body but carry out their function in another part of the body. State the name of this group of chemicals. [1]

....

(iv) **Image 3.1** shows that a fall in the level of glucose in the blood brings about a series of responses that result in its increase. State the name of this mechanism of control. [1]

....

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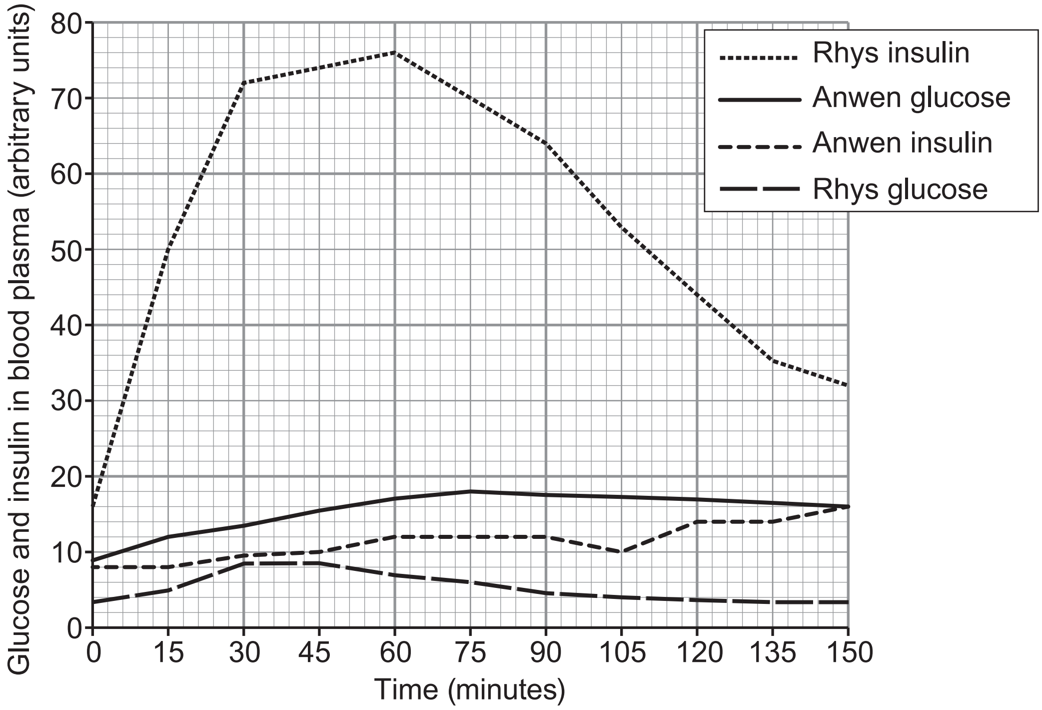
(b) Anwen and Rhys were each given an identical volume and concentration of glucose solution to drink. The concentration of glucose and of insulin in their blood was measured at regular intervals over the following 150 minutes. The results are shown in **Graph 3.2**.

Graph 3.2



(i) I. Describe **three** pieces of evidence from **Graph 3.2** which show that Anwen has diabetes. [3]

1. ....

....

2. ....

....

3. ....

....